

Instance Based Learning Vs Model Based Learning

Use Case

ML Model

Regression
Classification

Learning pattern

Instance Based

Model Based Technique

Learning

Memorizing

Generalizing

Instance Based

Model Based

Learning

Training
↓
Store Data
↓
Prediction Time

DATA
Op

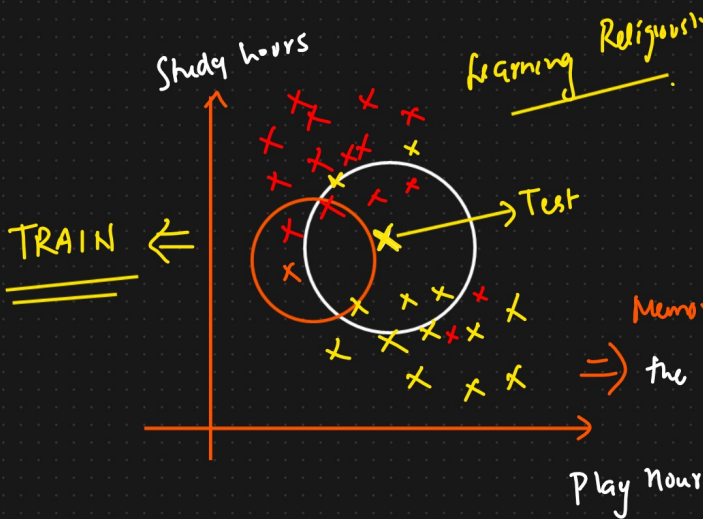
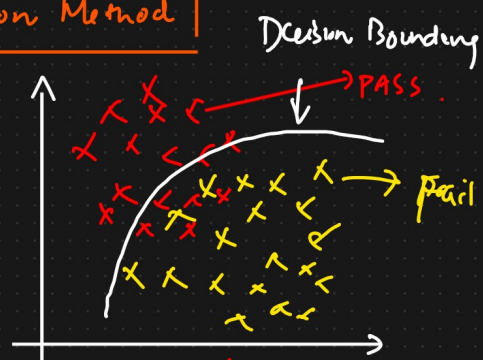
DATA
Pattern

Training
↓
Learn Pattern
↓
Create Model
↓
Prediction

Generalization Method

No. of play	No. of Study hours	Pass/Fail

Generalized Model



X → PASS
x → FAIL
Learn pattern of the data
Generalized Format

Instance Based Learning

Domain Expert

① KNN {K Nearest Neighbour}

$IP \rightarrow \boxed{\text{Model}} \rightarrow \text{Score}$
 $\downarrow$ New
 $\downarrow$ Model

$hs \rightarrow \text{Kbs or mb}$
 $\downarrow$

$pkc \Rightarrow \text{hard drive}$

$\Rightarrow$ Serialized format

Generalization
 Decision
 Boundaries

Usual/Conventional Machine Learning	Instance Based Learning
Prepare the data for model training ✓	Prepare the data for model training. No difference here ✓
Train model from training data to estimate model parameters i.e. <u>discover patterns</u>	Do not train model. Pattern discovery postponed until scoring query received
Store the model in suitable form	There is no model to store
Generalize the rules in form of model, even before scoring instance is seen	No generalization before scoring. Only generalize for each scoring instance individually as and when seen
Predict for unseen scoring instance using model	Predict for unseen scoring instance using training data directly
Can throw away input/training data after model training	Input/training data must be kept since each query uses part or full set of training observations
Requires a known model form	May not have explicit model form
Storing models generally requires less storage	Storing training data generally requires more storage
Scoring for new instance is generally fast	Storing for new instance may be slow

$\uparrow$ Yes or No

$IP \xrightarrow{pkc} \boxed{\text{Model}} \rightarrow O/P$